BUSINESS DATA ANALYSIS

Retail Sales & Profitability Analysis

Sample Superstore Business Dataset

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Data Analysis Project

Tools: Excel • SQL • Power BI / Data Visualization

Analysis period: 2014–2017

Project focus

Understanding sales growth, profitability, customer segments, regional performance, and product-level opportunities.

1. Executive Summary

This project examines a public business dataset containing retail transaction information such as order dates, customers, regions, product categories, discounts, sales, quantities and profit. The purpose is not only to report what happened, but to understand where the business is performing well and where sales may not be translating into profit.

The dataset contains 5,009 distinct orders, 793 customers and 37,873 units sold. Total sales are approximately $2,297,200.86, while total profit is $286,397.02, giving an overall profit margin of about 12.46%.

The analysis shows a clear long-term increase in sales between 2015 and 2017, but the results also show why sales alone are not enough to judge business performance. Technology generates the largest category-level profit, while Furniture produces comparatively low profit despite substantial sales. At the sub-category level, Tables and Bookcases stand out as loss-making areas, whereas Copiers, Phones and Accessories contribute strongly to profit.

2. Business KPI Snapshot

Total Sales: $2,297,200.86

Total Profit: $286,397.02

Profit Margin: 12.46%

Distinct Orders: 5,009

Customers: 793

Units Sold: 37,873

3. Dataset and Business Context

The project uses the Sample Superstore dataset published through Tableau's sample-data resources. It is designed as a retail business training dataset rather than a confidential company database. That makes it useful for demonstrating an end-to-end analytics workflow while keeping the analysis reproducible.

Main fields used include Order Date, Ship Date, Ship Mode, Customer ID, Segment, Region, State, City, Category, Sub-Category, Sales, Quantity, Discount and Profit.

4. Analysis Methodology

1. Understand the business question — Where are sales coming from? Which products create profit? Which areas need attention?

2. Validate the data — Check columns, data types, date fields, missing values, duplicates and unusual values.

3. Create useful measures — Total sales, total profit, profit margin, order count, customer count, quantity and year-level metrics.

4. Segment the data — Break results down by year, region, category, sub-category and customer segment.

5. Visualize patterns — Use charts so changes, comparisons and outliers are easier to understand.

6. Convert findings into business actions — Connect the numbers with possible actions such as reviewing discount levels, product pricing and regional strategy.

5. Data Preparation and Quality Checks

Before analysis, a business analyst should not immediately start building charts. The raw table needs basic validation. Recommended checks include column structure, date fields, numeric fields, duplicate rows, missing values and negative profit.

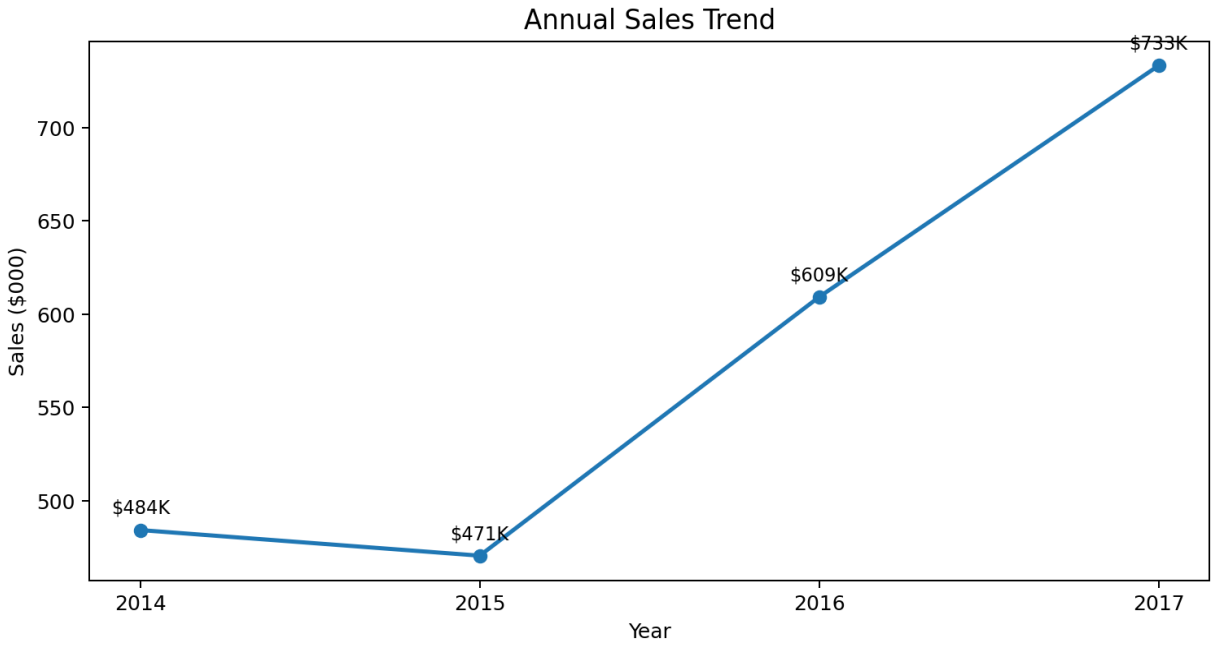
An important point is that negative profit should not automatically be deleted. In a retail dataset, a negative-profit transaction can represent a genuine loss caused by discounting, product economics or order-level circumstances. Removing those rows would make the business look healthier than it actually is.

6. Sales Trend Over Time

Annual sales were approximately $484.25K in 2014, fell slightly to $470.53K in 2015, then increased to $609.21K in 2016 and $733.22K in 2017.

From 2015 to 2017, annual sales increased by about 55.8%. This suggests that the business expanded its sales volume over the later part of the period.

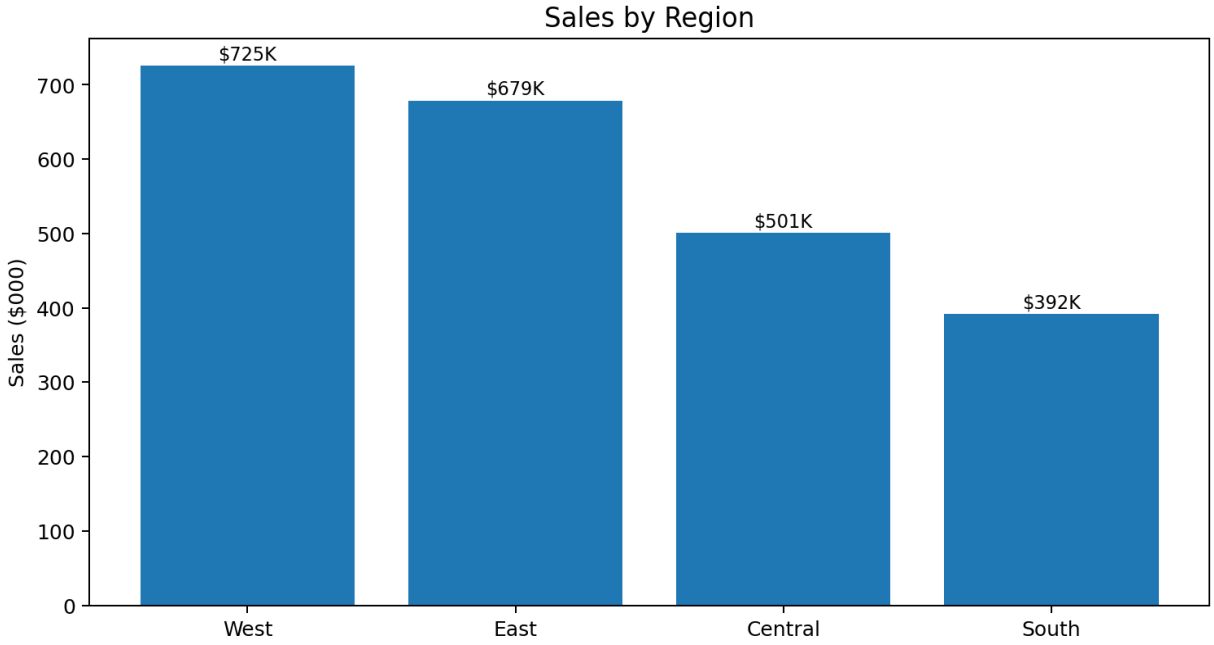
However, a growing sales line should be interpreted together with profit. Revenue growth is useful only when the business retains a reasonable share of that revenue after discounts and other costs.



7. Regional Performance

The four major regions are West, East, Central and South. West has the highest sales total and the highest total profit. East is the second-largest sales contributor. Central has substantially lower profit than West and East, while South has the lowest sales total in this comparison.

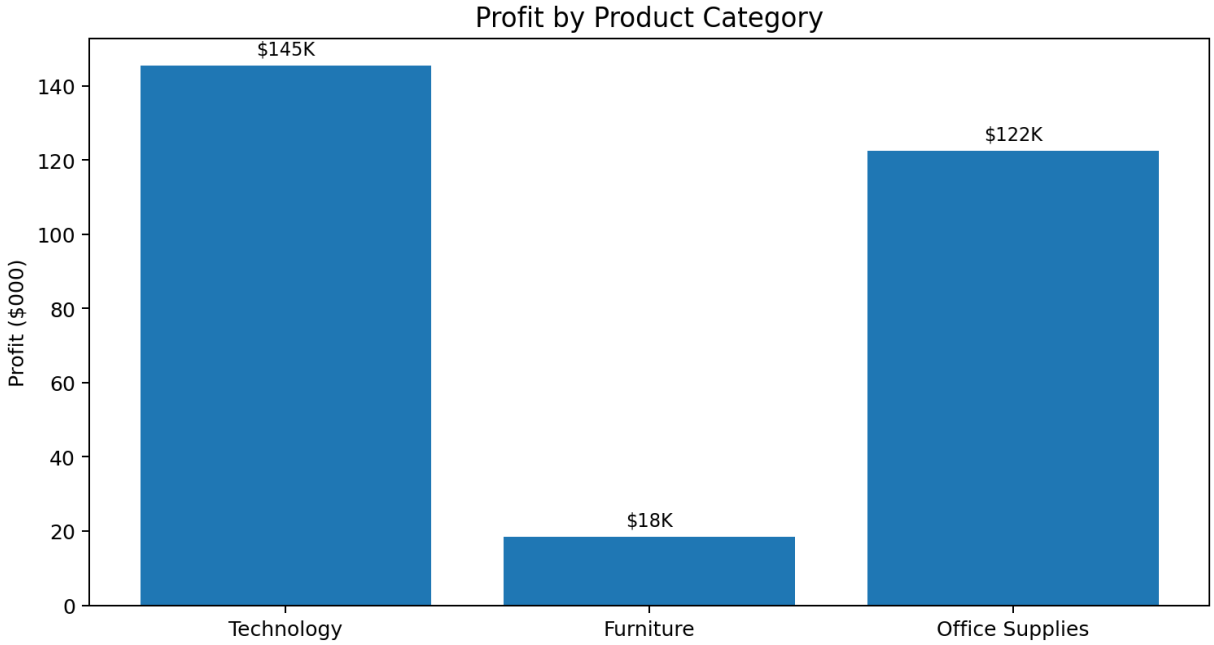
A useful next step for a real business would be to examine state-level performance inside each region rather than treating an entire region as one market.



8. Product Category Analysis

Technology is the strongest category in terms of profit, generating about $145.5K. Office Supplies follows with about $122.5K. Furniture is the major warning area: it generates roughly $742.0K in sales but only about $18.5K in profit.

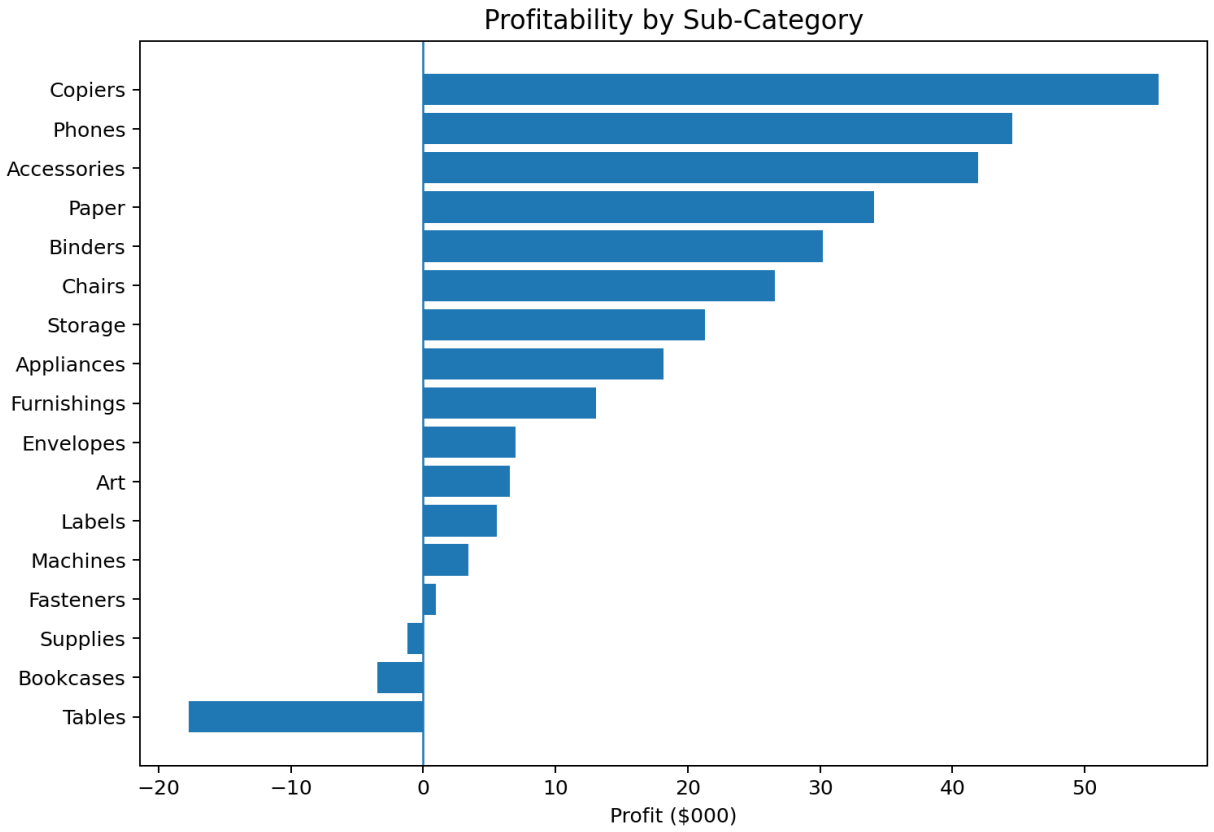
This difference is important because a category can look successful on a sales dashboard while creating very little economic value. For management, category profitability is therefore more informative than revenue ranking alone.



9. Sub-Category Profitability

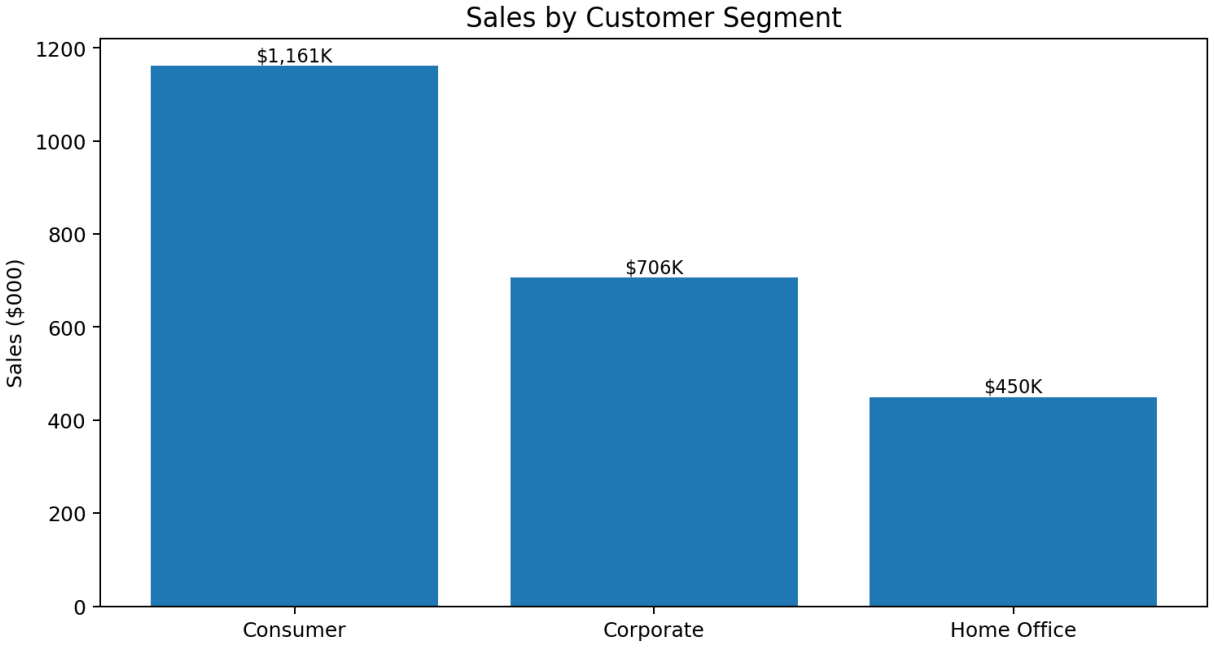
The sub-category view gives a more detailed picture. Copiers, Phones and Accessories are among the strongest positive contributors. At the other end, Tables shows the largest negative profit in the dataset, followed by Bookcases.

The practical question is not simply whether a sub-category has low profit. The next investigation should ask why. Possible drivers include high discount rates, product cost, pricing, shipping cost, product mix and individual product performance.



10. Customer Segment Analysis

Consumer is the largest segment by sales, followed by Corporate and Home Office. This tells the business where most revenue is currently coming from, but it does not mean every segment should be treated identically. Segment-level profit margin and repeat purchasing would be useful follow-up metrics.



11. What the Numbers Say Together

When the major findings are viewed together, the dataset tells a more useful story than any single chart. Sales are growing over time, but profitability is uneven across products. The West and East regions contribute a large share of

revenue, while Technology and Office Supplies provide much stronger profit than Furniture.

This creates a classic business-analysis situation: the company does not necessarily need to sell more of everything. It needs to understand which sales create healthy profit and which sales consume discounts or costs without sufficient return.

12. Business Questions Answered

Where are sales strongest? — The West region has the highest sales total, followed by East.

Which category creates the most profit? — Technology has the highest total profit among the three major categories.

Is the highest-sales category automatically the most profitable? — No. Furniture has high sales but much lower profit.

Which products deserve deeper review? — Loss-making sub-categories such as Tables and Bookcases deserve investigation.

Is the business growing? — Annual sales show a strong increase from 2015 through 2017.

Which customer segment is largest? — Consumer contributes the largest sales total.

13. Business Recommendations

Review discounting on low-profit products. Instead of applying broad discounts, review discount levels at sub-category and product level.

Protect strong technology categories. Technology is a major profit contributor, so stock availability, pricing and product mix should be monitored.

Investigate Furniture economics. Furniture should be analyzed at sub-category and product level, especially Tables and Bookcases.

Use region-specific strategies. West and East have large sales contributions, while Central and South should be studied at state level.

Track profit margin with every sales report. Sales, profit and margin should appear together in management reporting.

Move from descriptive to diagnostic analysis. After identifying a weak category, drill into discount, product, state, customer and shipping dimensions.

14. Suggested Management Dashboard

Top KPI cards: Total Sales | Total Profit | Profit Margin | Orders | Customers

Line chart: Sales by Year / Month

Bar chart: Sales and Profit by Region

Bar chart: Profit by Category

Horizontal bar: Profit by Sub-Category

Segment chart: Sales by Customer Segment

Filters: Year, Region, Category, Segment, Ship Mode

The dashboard should avoid too many visual elements on one page. A manager should be able to identify the main performance issue quickly and then use filters to investigate it.

15. Example SQL Analysis Queries

```
SELECT SUM(Sales) AS Total_Sales, SUM(Profit) AS Total_Profit FROM Superstore;
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```
SELECT Region, SUM(Sales) AS Total_Sales, SUM(Profit) AS Total_Profit FROM Superstore GROUP BY Region ORDER BY Total_Sales DESC;

SELECT Category, SUM(Sales) AS Total_Sales, SUM(Profit) AS Total_Profit FROM Superstore GROUP BY Category ORDER BY Total_Profit DESC;

SELECT [Sub-Category], SUM(Sales) AS Total_Sales, SUM(Profit) AS Total_Profit FROM Superstore GROUP BY [Sub-Category] HAVING SUM(Profit) < 0 ORDER BY Total_Profit;

SELECT YEAR([Order Date]) AS Order_Year, SUM(Sales) AS Total_Sales FROM Superstore GROUP BY YEAR([Order Date]) ORDER BY Order_Year;
```

16. Excel Workflow for Reproducing the Project

Import the CSV/Excel dataset into Excel.

Check data types and blank cells.

Create PivotTables for Region, Category, Sub-Category and Segment.

Create calculated fields/measures for Sales, Profit and Profit Margin.

Insert PivotCharts for the major comparisons.

Add Year/Month fields from Order Date.

Use slicers for Year, Region, Category and Segment.

Build a one-page dashboard and keep a separate sheet for detailed analysis.

17. Limitations and Next Steps

This project is intended for business-analysis practice. The Sample Superstore dataset is a public training dataset representing a fictional retail business, so the results should not be interpreted as evidence about a real company.

The dataset does not provide complete cost accounting; customer lifetime value and repeat-purchase behavior are not deeply analyzed; the project focuses on descriptive and diagnostic analysis rather than a production forecast; and a real business would ideally combine sales data with inventory, marketing, returns, logistics and cost data.

For a second version, useful additions would include monthly seasonality, state-level maps, discount-vs-profit scatter analysis, shipping-time analysis and a Power BI dashboard connected to cleaned data.

18. Conclusion

The main lesson from this analysis is that business data analysis is not just about making charts. The useful part is connecting the numbers to a business question. The dataset shows a growing sales trend, strong regional differences and large differences in profitability between product categories and sub-categories.

The most important management takeaway is to look beyond revenue. Technology and Office Supplies demonstrate stronger profit contribution, while parts of Furniture require closer investigation. Similarly, regional performance should be understood at a more detailed level before making operational decisions.

Overall, this project follows a simple analyst workflow: understand the data → clean and validate → calculate KPIs → compare segments → visualize patterns → explain what the patterns mean → recommend what should be investigated next.

19. References

Tableau Public – Sample Data, Superstore Sales: public Tableau sample-data resource.

Sample Superstore public dataset copies used for reproducible analysis and field verification.

Analysis calculations in this report are based on standard Sample Superstore aggregate values and derived ratios.

Source page: https://public.tableau.com/app/learn/sample-data?qt-overview_resources=1

20. Project Submission Checklist

- ■ Dataset identified and described
- ■ Business problem clearly stated
- ■ Data preparation and quality checks documented
- ■ KPIs calculated
- ■ Time trend analyzed
- ■ Regional performance analyzed
- ■ Category and sub-category profitability analyzed
- ■ Customer segment analysis included
- ■ Charts included
- ■ Business recommendations connected to findings
- ■ SQL examples included
- ■ Limitations documented